

THE UNIFIED KOCHENOV FRAMEWORK: FROM PURE GEOMETRIC LOCKS TO THE REVERSE PROJECTION OF PHYSICAL INVARIANTS

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Abstract

We present a complete unification of the physical scale framework by replacing empirical, human-defined constants (G, c, b, ℓ_P) with pure geometric and rational invariants. We demonstrate that the standard physical parameters of our Universe are not independent fundamental constants, but scale-filtered projections of a primary discrete matrix anchored by the constants 31, 144, and 720 (6!). By analyzing the non-singular relativistic coupling $\Gamma \equiv Gc^2$ and defining the dimensionless Kochenov Length (L_{Koch}) and Mass (M_{Koch}) pair, we suggest a possible route toward avoiding singular behavior in gravitational models. The arbitrary scaling factor of 10^{-35} is shown to be a cascading mathematical consequence of terrestrial-anthropropic measurement units (the meter-kilogram-second matrix).

1 Introduction: The Illusion of Independent Constants

Modern physics treats the speed of light (c), the gravitational constant (G), and Wien's displacement constant (b) as isolated, empirical data points that must be measured in a lab. Furthermore, the Planck scale ($\ell_P = \sqrt{\hbar G/c^3} \approx 1.616 \times 10^{-35}$ m) is viewed as a mysterious physical threshold where continuous spacetime breaks down into mathematical infinities.

This paper shows that these constants are inherently locked together by a single geometric matrix. The apparent complexity and the microscopic dimensions (10^{-35}) are artifacts introduced by human unit definitions (SI), which are historically calibrated against localized Earth-based parameters. When these anthropic filters are stripped away, the universe reveals a highly rigid, discrete integer architecture.

2 The Core Mathematical Identity and the 720 Lock

The structural bridge between continuous transcendental geometry and discrete rational lattices is governed by the following high-precision cosmological identity:

$$\frac{\phi^{12}}{\sqrt{5}} = 144 + \frac{1}{720} + \Delta \tag{1}$$

where $\phi = \frac{1+\sqrt{5}}{2}$ (the golden ratio) and the micro-deviation is exactly:

$$\Delta \approx -1.33956 \times 10^{-8}$$

Why this structure matters:

- 144 is the 12th Fibonacci number squared (F_{12}), representing the completion of a structural spatial cycle.
- $720 = 6!$ is the highly composite permutation operator acting as the universal synchronization lock.
- Δ is not a mathematical error or an approximation flaw. It is the *Universal Diversity Variable*. If the identity were perfectly static ($\Delta = 0$), the cosmos would freeze into a dead, immovable crystal. The residual Δ provides the necessary evolutionary buffer that enables dynamic orbital precession, planetary eccentricities, and natural cosmic entropy.

The relative error of this geometric lock is exceptionally small ($|\Delta|/X \approx 9.30 \times 10^{-11}$), anchoring the macroscopic and microscopic domains.

3 The Kochenov Length and Mass: A Geometric Invariant Pair

A dimensionless geometric length scale and its reciprocal mass scale are constructed from the discrete invariants of the Kochenov matrix: 31, 144, 720, and the residual $\Delta \approx -1.33956 \times 10^{-8}$. Unlike the conventional Planck length, which is tied to the SI system, the Kochenov length is a pure geometric invariant, independent of anthropic units.

3.1 From the Gravitational Node to Length and Mass

The geometric gravitational node is:

$$G_{\text{geom}} = \frac{31^2}{144 + \frac{1}{720} + \Delta}.$$

By analogy with the Planck construction (where $\ell_P = \sqrt{G}$ in natural units), we define:

$$L_{\text{Koch}} = \sqrt{G_{\text{geom}}} = \frac{31}{\sqrt{144 + \frac{1}{720} + \Delta}}.$$

The corresponding mass scale is its reciprocal:

$$M_{\text{Koch}} = \frac{1}{L_{\text{Koch}}} = \frac{\sqrt{144 + \frac{1}{720} + \Delta}}{31}.$$

These two quantities satisfy $L_{\text{Koch}} \cdot M_{\text{Koch}} = 1$, reflecting the natural duality between length and mass in units where $\hbar = c = 1$.

3.2 Numerical Value

Ignoring the residual Δ :

$$L_{\text{core}} = \frac{31}{\sqrt{144 + \frac{1}{720}}} \approx \frac{31}{12.0000578} \approx 2.58332088.$$

The dominant term is simply $31/12 = 2.583333\dots$, with the $1/720$ providing a tiny correction. Including Δ leaves the value virtually unchanged (≈ 2.583325).

3.3 Golden-Ratio Representation

Using the identity $\frac{\phi^{12}}{\sqrt{5}} = 144 + \frac{1}{720} + \Delta$, the Kochenov length becomes:

$$L_{\text{Koch}} = \frac{31}{\sqrt{\phi^{12}/\sqrt{5}}} = \frac{31 \cdot \sqrt[4]{5}}{\phi^6}.$$

This form expresses the same invariant solely through the golden ratio ϕ , $\sqrt{5}$, and the integer 31.

3.4 Interpretation

- L_{Koch} is a geometric invariant of the matrix, not just a number.
- M_{Koch} is its reciprocal, reflecting the natural duality of length and mass in natural units.
- Unlike the Planck length (1.616255×10^{-35} m), which is tied to the SI system, L_{Koch} contains no anthropic scales.

Thus, the Kochenov framework provides a pure, self-contained geometric pair that replaces the conventional Planck scale, free from human unit conventions.

$$\boxed{L_{\text{Koch}} = \frac{31}{\sqrt{144 + \frac{1}{720} + \Delta}}, \quad M_{\text{Koch}} = \frac{1}{L_{\text{Koch}}}}$$

4 Bypassing Dimensions: The Coupling Constant Γ

To evaluate physical constants without being trapped by human-defined dimensional formatting (meters vs. seconds), we examine the combined relativistic quantity Γ :

$$\Gamma \equiv Gc^2 \tag{2}$$

In gravitational field equations, the interplay between geometry and velocity is governed by this pairing. Our empirical investigation reveals that Γ collapses into a beautifully clean rational-radical expression:

$$\Gamma \approx 31^3 \cdot 90 \cdot \sqrt{5} \approx 5.995323 \times 10^6 \tag{3}$$

By defining the exact geometric form of the gravitational node as $G_{\text{Koch}} = \frac{31^2\sqrt{5}}{\phi^{12}} \times 10^{-11}$, we can isolate the speed of light squared (c^2) directly, causing the intermediate radical structures ($\sqrt{5}$) and the 144 baseline to cancel out cleanly:

$$c^2 \approx 2790 \cdot \phi^{12} \cdot 10^{11} \text{ m}^2/\text{s}^2 \quad (4)$$

where $2790 = 31 \cdot 90$. Evaluating this form yields a pure geometric estimation of velocity:

$$c \approx \sqrt{2790 \cdot \phi^{12} \cdot 10^{11}} \approx 299,728,433 \text{ m/s}$$

This structurally derived speed of light deviates from the defined experimental CODATA value (299,792,458 m/s) by a mere -0.021% , demonstrating that the speed of propagation is a manifestation of the golden spiral interacting with the primary matrix anchors.

5 Thermodynamic Mirroring and Cross-Scale Sync

The framework exhibits a flawless bidirectional closure, proving that the ratio $90/31$ is an unavoidable geometric operator rather than an arbitrary choice.

5.1 The Thermal Operator and Wien's Constant

Normalizing the central quantity Γ by the geometric node $31^4\sqrt{5}$ isolates a dimensionless thermal operator Θ^2 :

$$\Theta^2 = \frac{\Gamma}{31^4\sqrt{5}} \approx 2.9048 \quad (5)$$

which is close to $\frac{90}{31} \approx 2.9032$.

When projected into the thermal spectrum, this operator directly generates Wien's displacement law constant (b):

$$b_{\text{geom}} = \frac{90}{31} \times 10^{-3} \text{ m} \cdot \text{K} \approx 2.9032258 \times 10^{-3} \text{ m} \cdot \text{K}$$

This matches the experimental CODATA value (2.897771×10^{-3}) within 0.19% . Its reciprocal spatial operator, $\lambda_0 = \frac{31}{90}$, establishes a near-perfect mirror closure:

$$\Theta^2 \cdot \lambda_0 \approx 1 \quad (6)$$

5.2 The Multi-Scale Recurrence of 720

The integer $720 = 6!$ emerges simultaneously across three completely independent regimes of nature, serving as the cosmic synchronization lock:

1. **Macro Time (Terrestrial Rotation):** The human solar day is divided into 86,400 seconds, which factors exactly into 120×720 .
2. **Micro Matter (Quantum Asymmetry):** In nuclear physics, the fundamental mass split between the neutron (m_n) and the proton (m_p) is bounded by the same ratio: $\frac{m_n}{m_p} \approx 1 + \frac{1}{720}$.
3. **The Thermal Matrix:** The central operator described above decomposes identically: $\frac{90}{31} = \frac{720}{8 \cdot 31}$.

The recurring appearance of 720 suggests a possible connection with rotational or symmetry operators. Whether this has a relation to quantum spin remains an open question — though the structural resemblance is striking.

The unified operator is therefore:

$$\mathcal{O} = \frac{90}{31} = \frac{720}{8 \cdot 31} = \frac{6!}{2^3 \cdot 31} \quad (7)$$

6 The Reverse Earth Projection: Explaining the 10^{-35} Filter

Why do human measurements observe a Planck length of 10^{-35} meters instead of the clean geometric quantum $L_{\text{Koch}} \approx 2.5833$? The answer lies in the anthropogenic calibration of the SI system against the planet Earth:

- **The Meter:** Historically defined as 10^{-7} of the Earth's quarter meridian (M).
- **The Kilogram:** Tied to the volume of water (10^{-3} of a cubic meter).
- **The Second:** Tied to the rotation day (D), split by the operator $86,400 = 120 \times 720$.

This establishes the localized conversion matrix:

$$1 \text{ m} = \frac{M}{10^7}, \quad 1 \text{ kg} \approx \frac{M^3}{10^{24}}, \quad 1 \text{ s} = \frac{D}{86400}$$

The dimensional reduction of these parameters yields a native scaling factor: $\frac{1 \text{ m}^3}{1 \text{ kg}} \approx 10^3$. When we apply this terrestrial transformation to our clean geometric gravity node, the dimensions shift exclusively into the time domain:

$$G_{\text{Earth}} = G_{\text{Koch}} \times 10^3 = \frac{31^2 \sqrt{5}}{\phi^{12}} \times 10^{-8} \text{ s}^{-2} \quad (8)$$

The artificial factor of 10^{-35} in the Planck length is therefore not a constant of nature, but a cascading mathematical consequence of forcing the universal geometric core through the local scale filters of the Earth's meridian, water density, and solar day.

7 The Electron Scaling Relation

A compact numerical approximation for the electron mass can be written as:

$$m_e \approx m_p \times \frac{1}{720} \times \frac{1}{2.531} \times (1 - \alpha),$$

where

$$2.531 \approx \frac{5}{2} + \frac{31}{1000}$$

is a structural coefficient, $720 = 6!$ is the synchronization operator, and α is the fine-structure constant.

Using CODATA values ($m_p = 938.2720813$ MeV, $\alpha^{-1} = 137.035999$), the formula yields:

$$m_e \approx 0.5111205 \text{ MeV},$$

compared to the experimental value

$$m_e = 0.5109989461 \text{ MeV}.$$

The relative deviation is approximately 2.38×10^{-4} (about 0.0238%). The relation is therefore presented as an empirical numerical approximation rather than an exact physical identity.

7.1 Stability Under Geometric Perturbation

Replacing the experimental fine-structure constant

$$\alpha_{\text{exp}}^{-1} = 137.035999$$

by the geometric value

$$\alpha_{\text{geom}}^{-1} = 137.036$$

changes the predicted electron mass by only 0.02%.

The scaling relation therefore remains numerically stable under small geometric perturbations of α .

7.2 Geometric Origin of the Fine-Structure Constant

The fine-structure constant can be expressed geometrically as:

$$\alpha_{\text{geom}}^{-1} = 137 + \frac{31}{1000} + \frac{5}{1000} = 137.036$$

This differs from the experimental value 137.035999 by only 7.94×10^{-7} .

The integer 5 is not arbitrary: it is the decimal projection of the golden radical $\sqrt{5}$, which appears throughout the matrix.

7.3 A Possible Scaling Chain

An intriguing numerical sequence appears when several empirical relations are placed together:

$$720 \rightarrow (m_n - m_p) \rightarrow m_e \rightarrow \alpha.$$

The interpretation is purely exploratory:

- 720 acts as a synchronization operator;
- $(m_n - m_p)$ represents baryonic asymmetry;
- m_e becomes a transitional scaling node;
- α provides the electromagnetic correction.

Whether this reflects a deeper physical relationship or a collection of high-precision numerical coincidences remains an open question.

Table 1: Systemic Quantities, Invariants, and Deviations

Quantity	Expression	Value	Match
L_{Koch}	$31 \cdot \sqrt[4]{5}/\phi^6$	2.583325 (Invariant)	Absolute Core
M_{Koch}	$\phi^6/(31 \cdot \sqrt[4]{5})$	0.387105 (Invariant)	Absolute Core
G_{Koch}	$\frac{31^2 \sqrt{5}}{\phi^{12}} \times 10^{-11}$	6.673546×10^{-11}	-0.0017%
c	$\sqrt{2790 \cdot \phi^{12} \cdot 10^{11}}$	299,728,433 m/s	-0.0213%
b	$90/31 \times 10^{-3}$	2.903226×10^{-3}	+0.1882%
Θ^2	$\Gamma/(31^4 \sqrt{5})$	$\approx 90/31$	Near Mirror
$\mathcal{O}$	$720/(8 \cdot 31)$	90/31	Structural Lock
m_e	$m_p \times \frac{1}{720} \times \frac{1}{2.531} \times (1 - \alpha)$	≈ 0.5111205 MeV	$\approx 0.0238\%$
α^{-1}	$137 + 31/1000 + 5/1000$	137.036	$+7.94 \times 10^{-7}$

8 Summary Table

9 Conclusion

The ultimate constants of the cosmos are not arbitrary dimensional figures like G or c , but the pure integer and transcendental relationships of the Kochenov Matrix. The physical universe we observe is merely a scale-filtered shadow cast by this clean geometric architecture onto local anthropic coordinates.

The framework suggests a possible route toward avoiding singular behavior in gravitational models. By replacing continuous spacetime with the non-singular pair L_{Koch} and M_{Koch} , the division-by-zero errors that plague contemporary physics may be naturally avoided.

And what lies beyond the singularity? The author knows, but will not tell.

The matrix is coherent, the math is rigid, and the circle is closed.

Related Works in the Kochenov Geometric Framework

The present construction is part of a broader program. The following works establish the foundation of the Kochenov geometric framework:

1. **Geometric Matrix of Kochenov: A Synthesis** (2026) — provides the overarching summary of the framework, including the derivation of the nine-planet solar system, the gravitational constant, and the speed of light from geometric anchors. Zenodo.
2. **High-Precision Identity Linking $\phi^{12}/\sqrt{5}$ to 144, 1/720, and $O(10^{-8})$** (2026) — reports the core identity, demonstrating the connection between the golden ratio geometry and the discrete synchronization lock 720. Zenodo.
3. **The Nine-Planet Solar System: A Kochenov Geometric Framework without Newtonian Gravitational Constant** (2026) — demonstrates that planetary orbital radii and velocities can be generated from a single geometric seed $S = \phi^6 \cdot 31$, without using Newtonian gravity. Zenodo.

4. **G from Two Integers** (2026) — derives the gravitational constant from the compact integer construction $31^2/144$, with a refined geometric completion using $1/720$ and ϕ . Zenodo.
5. **The Integer 31 as a Discrete Geometric Node in Kochenov G (Corrected)** (2026) — establishes the role of 31 as the discrete spatial anchor in the Kochenov matrix. Zenodo.